

STUDENT ENERGY CONSULTING GROUP



AN ECONOMIC ANALYSIS OF A PHOTOVOLTAIC SYSTEM FOR THE UNIVERSITY OF ILLINOIS AT CHICAGO'S ENGINEERING INNOVATION BUILDING

Zainab Ahmed, Diana Awimrin,
Leena Daniel, Ariel Maret, and
Sherin Skariah

STUDENT ENERGY CONSULTING GROUP

TECHNICAL REPORT

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Executive Summary

Last semester, the University of Illinois at Chicago's (UIC) East Campus lost power twice. In the previous Chemical Engineering Building (CEB), the building lost power twice in one year. Now that the Chemical Engineering Department has moved into the Engineering Innovation Building (EIB) on East Campus, the security of electricity is even more important. In the past, when CEB experienced a power outage, students and faculty were evacuated from the building due to the potential health hazards. The evacuation was necessary because the ventilation system uses electricity to remove hazardous fumes from the research labs. Without the ventilation system, the fumes stay in the building. Typically, power has been restored to the building within one day, and students and faculty are allowed to return the following day. It will be assumed that 3 blackouts will occur each year that will last for a maximum of 6 hours.

With the evacuation of students and faculty, the Chemical Engineering Department loses valuable research time and instructional time. Additionally, if a power outage were to occur, data from experiments could be lost and might cause equipment to be restarted.

Currently, there are 469 photovoltaic panels on the roof of Lincoln and Douglas Hall that were installed in 2010, and are currently unused. The goal of this project is to provide an economic analysis for re-installing the unused photovoltaic panels on the roof of EIB. The energy generated by the panels would be stored in a battery. In order to reduce electricity costs for UIC, any additional energy would be used to power EIB and create Renewable Energy Credits (RECs). As part of the economic analysis, other renewable and non-renewable sources of energy will be investigated. Cost information will be calculated based on the current year, 2019, for a 15 year period, in Chicago, IL.

Introduction

Photovoltaic Cells at Lincoln and Douglas Hall

Currently, there are 469 photovoltaic panels on the roof of Lincoln and Douglas Hall that were installed in 2010, and are currently unused. On October 18th, 2019, the group inspected the solar panels on the roof with Marwan Chehade, Superintendent of Buildings and Grounds. According to Marwan, the solar panels are not in use because the employee responsible for the panels retired without passing on operational knowledge. ^[1] The size of the solar panels are 63 inches x 37 inches and manufactured by UP Solar.

Mawan suggested that we investigate the stability of the roof. Special equipment to mount and hold the solar panels at an angle are needed. In order to look closer at the design of the roof and electrical layout of Lincoln and Douglas Hall, the Office of Capital Planning and Project Management was contacted. Patricia Pac, the Associate Director of the Office of Capital Planning and Project Management supplied electrical, mechanical, and architectural drawings of Lincoln and Douglas Hall. See Appendix A for drawings. The solar panels on the roof are at an angle of 10 degrees. The array for Douglas Hall is 50 kW and the array for Douglas Hall is 51.5 kW for a total of 101.5 kW. Each panel has their own microinverter.

Additionally, the installation and cost of the panels was obtained from the Office of Capital Planning and Project Management. The individual costs of installation and the panels are not known because the bids were not itemized. Douglas Hall's solar panels were included in the initial project, but the solar project was later extended to Lincoln Hall. There are two different contracts for each of the buildings. The cost of electricity, in 2019, for Douglas Hall is \$12.21/W. The cost of electricity, in 2019, for Lincoln Hall is \$8.68/W. Both of these costs seem very high. In comparison, according to the U.S. Energy Information Administration, the cost of electricity in Illinois for commercial buildings in September 2019 is \$0.0885/kW. ^[2] One of the main reasons that the cost of electricity produced by the solar panels is expensive is that the cost of the panels and installation was very expensive in 2010. Lawrence Berkeley National Labs has published six reports regarding solar utility projects. The following figure below is from the 2018 report and demonstrates the difference in installation costs from 2010 until 2017. For

example, the median AC project drop \$3.00/W over the 7 year period. ^[3] Another factor impacting the cost of installation is location.

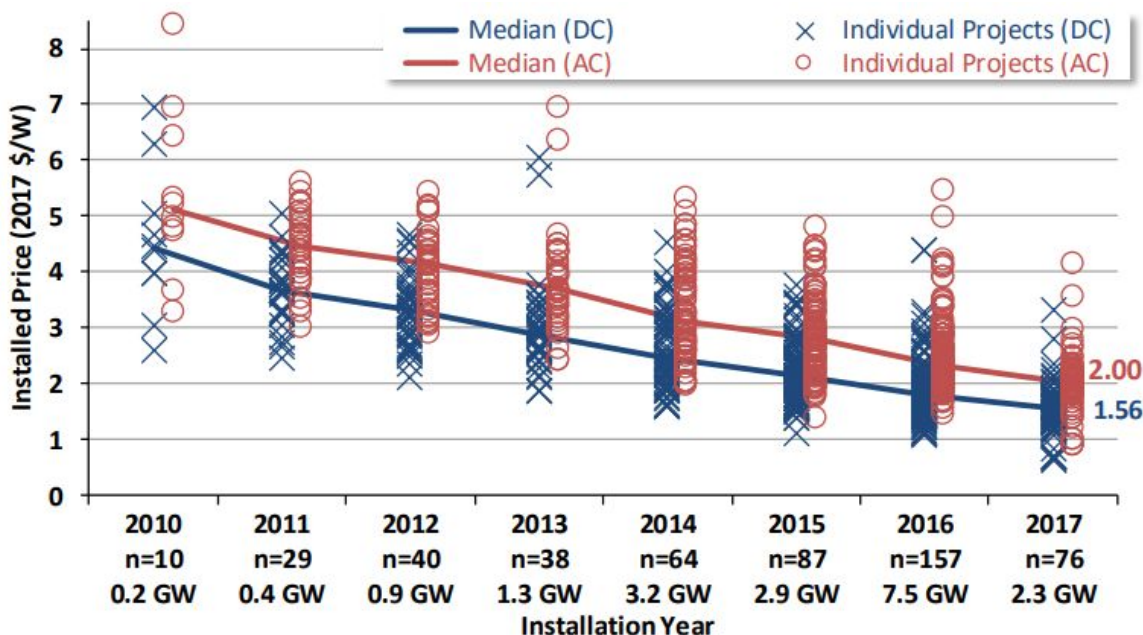


Figure 1: Installed Price of Utility-Scale PV and CPV Projects by Installation Year
Lawrence Berkeley National Laboratories ^[3]

Engineering Innovation Building

The Engineering Innovation Building (EIB) is 57,500 square feet and opened in July of 2019. It houses the entire Department of Chemical Engineering and some of the Department of Civil and Materials Engineering. ^[4] EIB is home to twelve labs.

Currently, EIB does not use renewable energy to generate electricity. According to Eric Luchetta-Stendel, the Senior Assistant Director at the Office of Capital Planning and Project Management, “...due to cost constraints at the time of design, the design team did not include renewable technologies.” ^[5]

Utility data was collected from Bradley Simmons, Utilities Deputy Director. Since EIB is a fairly new building, the data provided has been raw data. The utility data contains the energy usage for September 11, 2019 to November 11, 2019. The utility data is not broken down into

different categories. According to the U.S. Energy Information Administration, the majority of energy consumed by universities is heating and cooling and lighting.^[6] The utility data for EIB does include the electricity from chilled water (CW) and high temperature hot water (HTHW), but was not included as part of this analysis because they are produced at UIC's East Co-Generation Power Plant and then distributed to EIB.

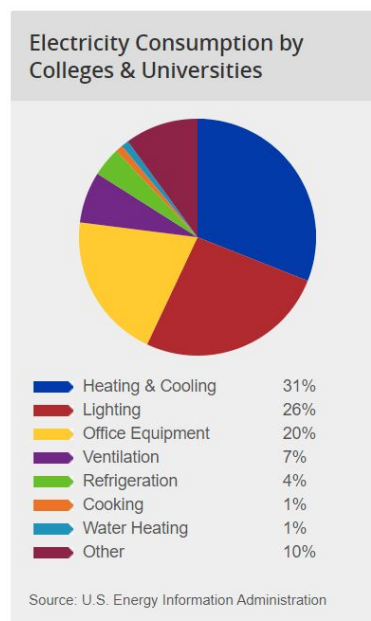


Figure 2: Electricity Consumed by Colleges and Universities ^[6]

Energy and Power Usage for EIB

If EIB did experience a power outage, the solution would need to supply enough energy to sustain EIB for a 6 hours of the day. To calculate the amount of electricity EIB uses, Simmons shared the raw data from the meter. The raw data includes energy usage over a period of 30 days from September 11, 2019 to October 11th 2019. Every fifteen minutes, the energy usage for the building is logged. Adding the total energy over a period of one day, the average energy usage is 1,589 kWh. The peak energy usage is 408 kW and occurred at 5pm.

Solution

Photovoltaic arrays were chosen because of its feasibility, given our location is in Chicago and is the most economical renewable energy for this case. Additionally, the total capital cost of the project was greatly reduced because new solar panels did not need to be purchased. An assumption of 3 blackouts per year that lasts for about 6 hours was made for the analysis. For safety purposes, energy will be stored into a battery that will supply power for 6 hours long.

Photovoltaic Analysis

In Chicago, the average peak hours of sun is around 4.4 hours.^[7] The current photovoltaic array produces 101.5 kW, as stated in the introduction, which means an average amount of energy produced by the array in one day is 446.6 kWh. There are 469 solar panels located in Lincoln and Douglas Hall, each has a width of 5.25ft and a length of 3.08ft, which means there is a necessity for a minimum area of approximately 7,591 ft². On the roof of EIB, the solar panels would be added to the service yard area. This area is approximately 9,856 ft², which means there is space to add all 469 solar panels. The amount of power to be supplied to EIB is 408kW for 6 hours. The cost of solar panels are \$350/kW.^[8] So, if UIC does decide to get new solar panels, then the cost for the solar panels would be around \$142,800. From using the solar panels which are already bought, UIC would be saving that much money on this project.

The battery is only used during the time of a power outage. The batteries that would be used would be 48V lithium-ion batteries, since there needs to be enough voltage to power the whole building. In this analysis, it is assumed that the outages last in the worst case, 6 hours; therefore, there would only need to be a battery that would supply 6 hours. In order for the battery to supply this amount, the battery would need to be sized properly with the depth discharge and the inefficiency factor included. The battery capacity needed for six hours is 3,084.48 kW. The batteries cost \$250/kWh^[8], hence the total cost of the battery would be around \$612,000. The batteries require no maintenance, once they've been properly installed.^[9]

Then, costs for an inverter, controller, and for installing and maintaining the whole system would need to be accounted for as well. The charge controller is used to regulate the charge that is going into the battery and prevents the battery from being overcharged and from overvoltage occurring.^[10] This will help the lifespan of the battery. The cost for charge controllers with 80A is \$1,347.16.^[11] To find the amount of amps in the system, the power, 408kW, was divided by the voltage of the battery, 48V, and therefore, the amps is 8500. There would need to be about 107 charge controllers and the total cost for charge controllers would be \$144,146.12. An inverter is used to convert the DC power from the battery to AC power. The cost of the inverter is \$250/kW,^[8] which means the total cost would be \$102,000. The installation cost for the whole system would be \$0.5/W^[8] and hence the total cost for installing the solar panels would be \$50,750. EIB Facility workers would be responsible for solar panel cleaning maintenance every three months. Therefore, the total costs upfront would be \$908,866. The total life cycle cost for 15 years would be \$2,744,896.

Renewable Energy Credit (REC) Demand

After the battery has been fully charged, the excess energy produced by the solar array would be used for loads in EIB. By using the excess energy, UIC will be avoiding costs downstream. Natural gas from utility providers are used to create electricity and steam in UIC's East Co-Generation Power Plant because it costs less to produce electricity on-site than it would be to buy all electricity from Com-ED. From there, the power plant distributes electricity, steam and chilled water to buildings on campus. This electricity is distributed through incoming lines supplied by ComED. A small amount of energy is purchased from ComED to power the generators within the co-generation plant.^[12]

Because campus is not fully powered by ComED, it makes UIC ineligible for many of the incentives that are offered from going solar. One type of incentive not offered by ComED but offered by the state of Illinois is the Renewable Energy Credit (RECs). A REC is a legal instrument that conveys to its owner, the right to claim the associated environmental attributes of its generating resources.^[13] One Renewable Energy Credit is issued when 1MWh of electricity is delivered to the grid. RECs can be bought and traded to meet a state's requirements for

renewable energy.^[14] According to Bradley Simmons, UIC currently buys RECs, which increases the overall cost of production of electricity. The solar array produces a total of 101kW per hour, which is 1/10th of a REC. Looking toward the future, Illinois' Energy Portfolio has mandated 25% of power generated from renewable energy by 2025. Currently, only 7% of power in Illinois comes from renewable energy.^[14] This means that most likely, UIC will need to purchase or produce RECs in the future. By directly using renewable energy, UIC can generate its own REC's therefore saving on the total cost of electricity.

Data on how much UIC spends on REC's or how much any commercial building spends was not available and so some assumptions about potential savings had to be made. EIB's daily average energy use is 1,589.80kW. Assuming that 25% of solar energy is required from daily electricity consumption to no longer need to buy REC's, 397.45kW of solar energy would need to be produced a day. One REC accounts for 1MW of solar energy and costs \$72.97.^[15] Currently, the value of RECs for commercial applications is equal to the amount of energy produced multiplied by the price set by Illinois Power Agency.^[14] The amount of energy produced by the solar panels roughly every 2.5 days (1000/397.45) accounts for 1MW of solar energy, i.e one REC no longer needing to be bought. Roughly 146 REC's (365/2.5) will no longer need to be bought over one year, amassing savings of \$10,653 a year. Over the next 15 years, a total of \$159,795 will be saved.

Alternative Solutions

Two different alternative solutions were chosen to be analyzed, one a renewable technology, the other a non-renewable technology. The alternative solutions that will be analyzed are a fuel cell technology and a diesel generator system, respectively.

Fuel Cell Analysis

Fuel cells have for the most part been used as mobile uses of energy. In this case, a fuel cell would be used as a stationary back-up power source for EIB when the grid fails. There are many articles of research that look into the possibility of having stationary fuel cells in use and

an assessment on their cost. The cost of fuel cells for the manufacturer is very dependent on the production size. This assessment will greatly depend on the seller's production size as a bigger production size will result in lower retail price. Thus, an assumption that the production size of fuel cells is 500 units for all sizes. For this plan, EIB has a peak power of 408 kW. For a fuel cell size of 1020 kW, ten 100 kW fuel cells and four 5kW fuel cells would be needed to fulfill the peak power requirement as an assumption of 40% efficiency is used, given statistics from 2015 as seen in Appendix B Figure 2.^[16] In 2007, a cost analysis was done on backup power using 5 kW fuel cells with the retail price of a 5 kW system with a 2 kg hydrogen cylinder and installation to be \$22,000.^[17] For a system capacity larger than 50 kW, Figure 3 seen in Appendix B is used to calculate the retail price per kW of a 100 kW system with hydrogen and installation.^[17] For EIB with a needed system of ten 100 kW and four 5 kW fuel cell systems, the retail cost with hydrogen supply and installation is calculated given the cost inflation index factors of 2010 as 167 and 2019 as 289.^[18] The upfront cost of the fuel cells, installation, and hydrogen would be \$2,262,212.11. This cost will definitely cover a 25-year lifespan of solar panels as the fuel cell can operate up to 5,000 hours.

The high retail price is initially assumed to be due to the high cost of platinum of \$900.40/troy ounce.^[19] Based on an 80 kW fuel cell system, platinum only accounts for \$8/kW.^[20] Thus, the total cost of platinum for EIB's fuel cell system is \$6,528, using the cost inflation index factor of 2012 as 200.^[18] Another high cost input to retail price is the cost of a 2 kg hydrogen storage for every 5 kW fuel cell will last 6 to 8 hours^[17] covers the assumption that a blackout will happen up to a length of time of 6 hours. With 408 kg compressed hydrogen needed for backup supply, it is difficult to find space for storage. Furthermore, another 408 kg of hydrogen needs to be on site for immediate supply once a blackout happens. This way, there is less opportunity for a blackout to happen without power delay. With hydrogen's price being \$8.00 per kg currently, the cost of hydrogen for EIB's fuel cell system with backup supply is \$6,528.^[21]

However, the issue then becomes how this compressed hydrogen can be stored in proximity. There is not enough storage space within EIB, so a hydrogen storage system will need

to be developed. For a 3 days' worth of hydrogen supply (18 kg), a storage facility would cost around \$50,000 in 2010.^[17] Thus, for EIB a hydrogen storage facility would cost approximately \$3,922,554.89; making the upfront cost of the entire system to be \$6,184,767. The main concern of choosing fuel cells is hydrogen storage cost, as the cost of storage is currently very high with the methods today, a fuel cell cannot be accepted to use as powering EIB.

Diesel Generator Analysis

Another, alternative solution that has been looked into is the diesel generator system. The analysis is carried out with the assumption that the generator and fuel tanks will not be replaced in the 15 year life-span left of the solar panels, though generators can last up to 25 years. In the United States, the electrical standards are 60Hz.^[22] Several Kohler generators were analyzed to determine which is the most cost efficient for the peak energy usage of 408 kW that EIB requires on a given day. The Kohler 500REOZJ, 60Hz met these requirements^[23] This generator's list price is at \$93,571. In an analysis by the National Renewable Energy Laboratory, a generator of 25-30 kW has an installation and permitting cost of \$24,000.^[24] When these values are scaled up for the 500REOZJ, the installation cost is \$480,000.

Next, the fuel consumption for the generator over 15 years is determined. The generator is only going to be running at 82% of its fuel consumption, so about 112.5 liter of fuel per hour will be consumed.^[23] In Illinois, the cost of diesel is \$0.79 per liter.^[25] Using the current fuel prices over 15 years, the cost of fuel is estimated to be \$23,996. Also, since the fuel needs to be delivered to UIC, that cost needs to be determined. The Energy Information Administration (EIA) estimates that fuel delivery accounts for 13% of the fuel cost.^[26] Assuming that fuel delivery is about 13% of the cost of fuel per liter, the total cost over 15 years is \$3,120. It's important to note that these costs are subject to change over the next 15 years. A storage tank is also needed in order to hold the diesel fuel for the generator. The amount of fuel consumed per year is 535 gallons. With that said, two 285 gallon storage tanks are to be purchased. The total cost of the tank is \$1,100 to hold the approximately 285 gallons of fuel.^[27] For two tanks, the cost will be \$2,200. Unlike photovoltaic systems, the diesel generator will require routine maintenance and replacement of oil, coolant, and belts. Using the 500REOZJ Engine Operating

Manual, the cost of parts over a 15-year period was calculated.^[28] The total cost of oil, coolant, and belt replacement is \$20,045.

Overall, the cost of the generator system over a 15 year period, using current cost estimates will equate to \$622,932. Though the cost of the generators is relatively low, deviating away from this solution would be best because the generators can get very noisy. Also, the purpose of this project is to generate energy through a renewable source, so that automatically makes the generator system less alluring.

Conclusion

The goal of this report is to analyze the feasibility of a solar photovoltaic system for UIC's Engineering Innovation Building (EIB) to produce and store electricity in the case of a power outage. Everyone must evacuate the building during power outages, disrupting student and faculty classes and research. The evacuation is necessary due to the shut-off of the ventilation system.

The most expensive solution to implement is the fuel cell which would cost 6.2 million US dollars. The total cost of the photovoltaic system for 15 years is 2.7 million US dollars, while the cost of a diesel generator is 623,000 US dollars. The most expensive component of the photovoltaic solution is the cost of the batteries. UIC can save additional money by implementing the photovoltaic solution. By producing its own renewable energy, UIC will save on number of REC's needed to be bought to be in accordance to Illinois legislation. These savings amount to roughly \$160,000 over the remaining life cycle of the solar panels of 15 years.

Even though the diesel generator is less expensive, we recommend that UIC continues to invest in renewable energy. Further utility analysis will need to be completed since EIB is a new building. Further investigation into the number of RECs and incentives would need to be completed to decrease the cost of the photovoltaic system. We also realize that due to emerging technology, the price of batteries should continue to decrease. At the most, we hope that our project has inspired other students, faculty, and utility personnel to continue taking interest in sustainability projects on campus.

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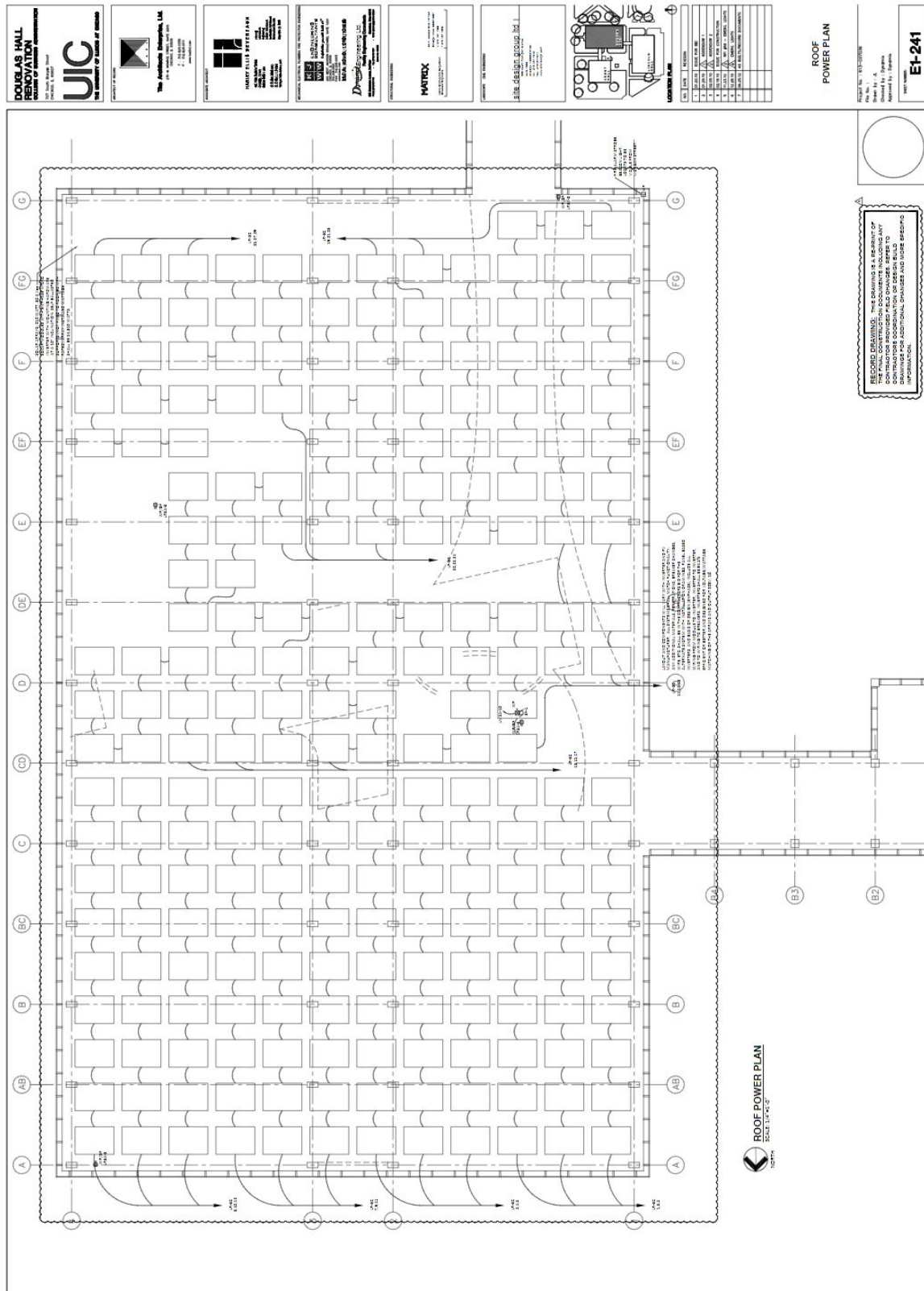
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Appendix A

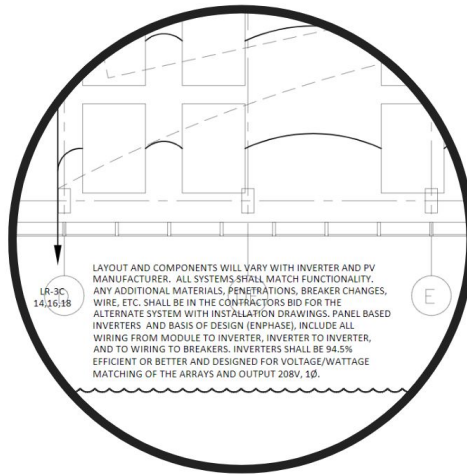


Figure AA-2: Enhancement of Electrical Wiring from Figure AA-1

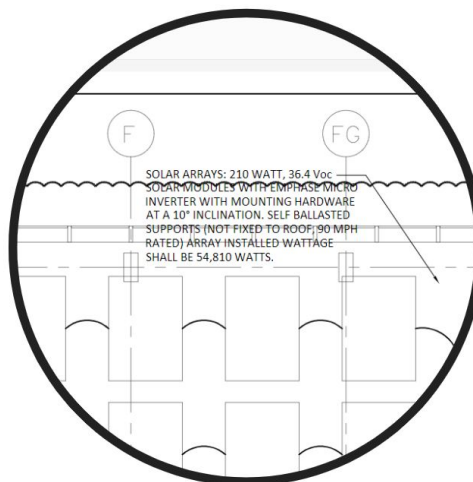
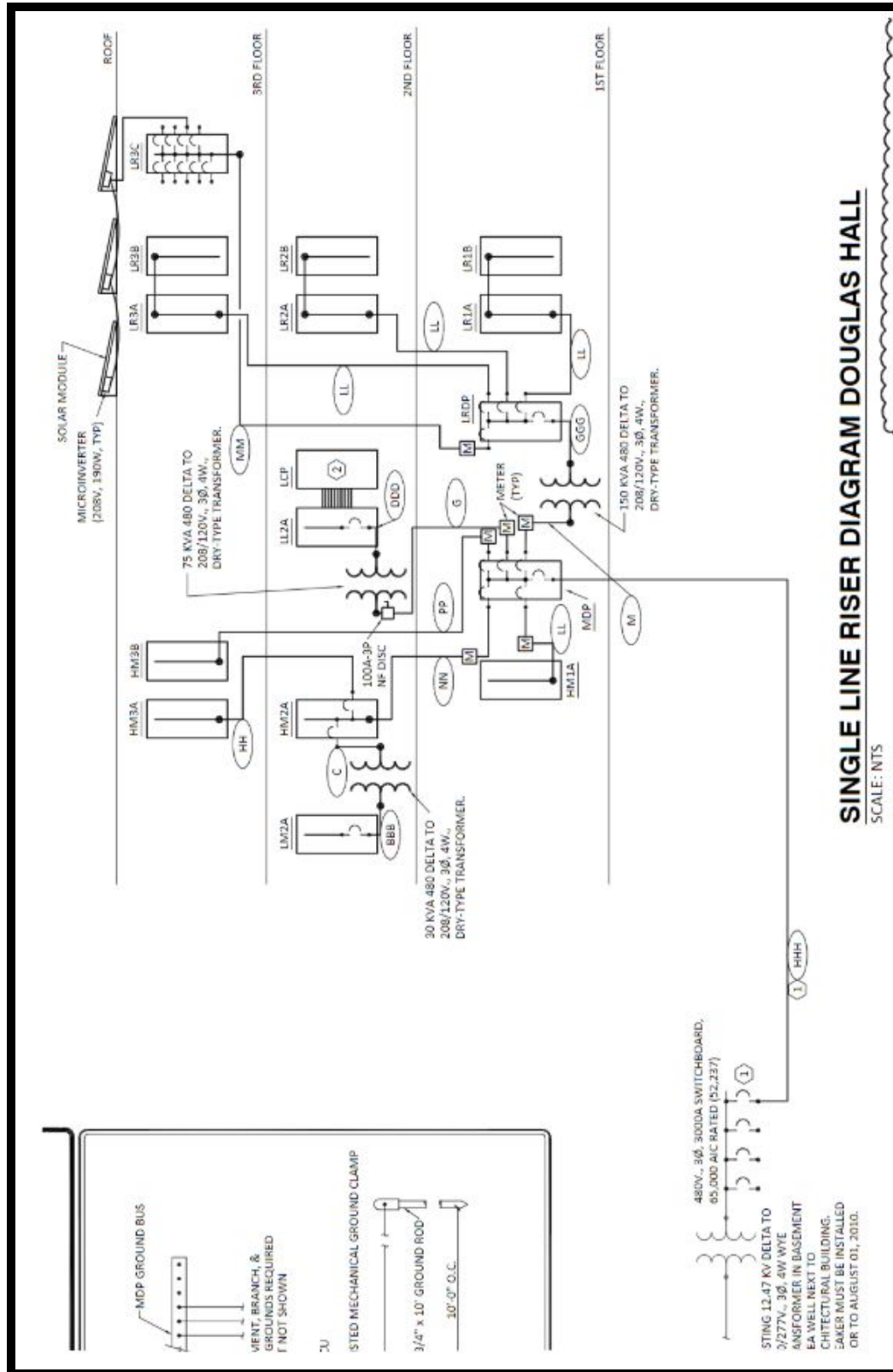


Figure AA-3: Enhancement of Solar Arrays from Figure AA-1

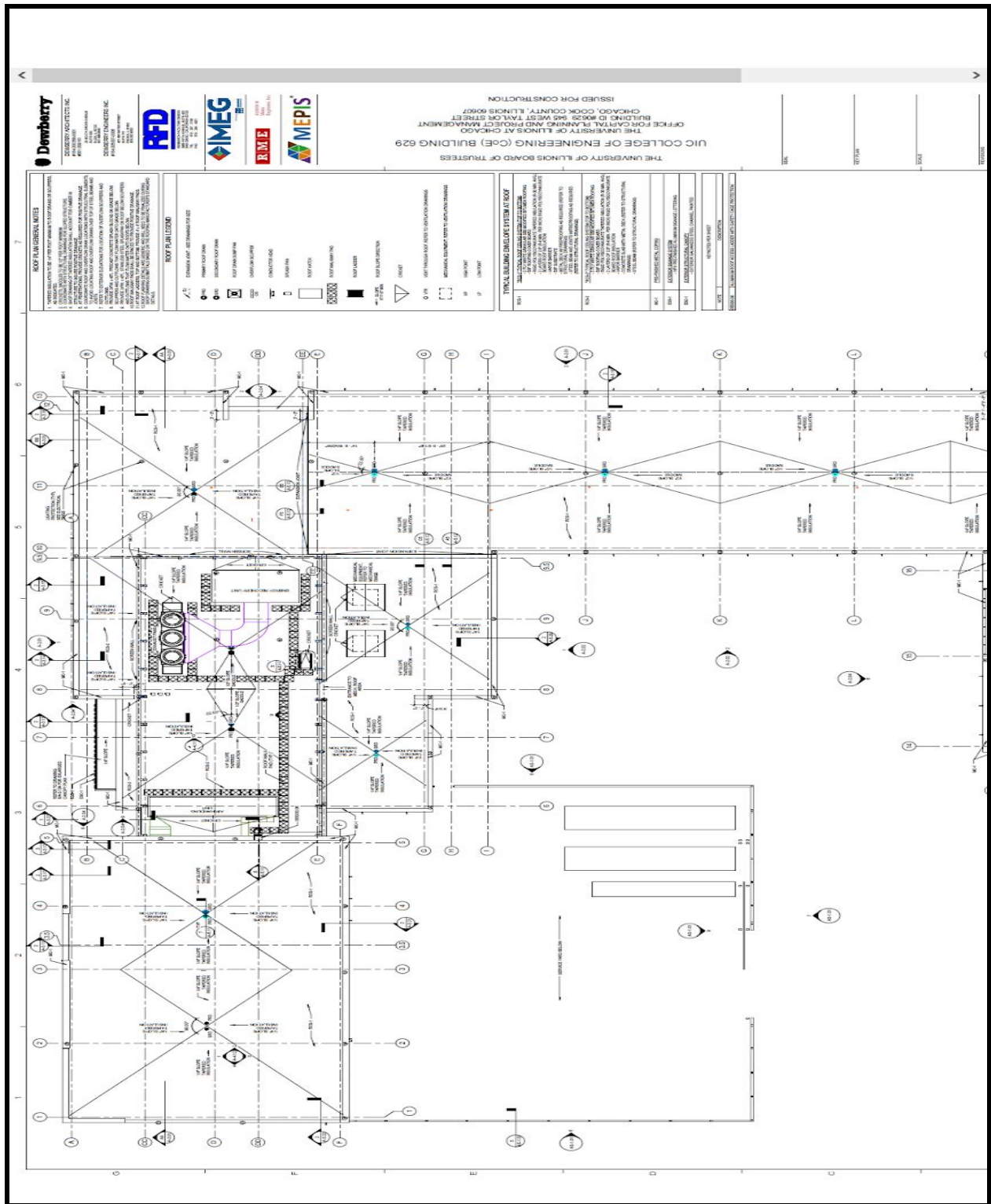
Appendix A



Appendix A

Project	Data Source	Data Provider	Estimate Date	System Specified	Base Construction Cost	Contractor Overhead and Fees	Total Cost ***	Total Cost Extended to November 2019 (from Turner Construction Index)	\$/Watt	Note
Douglas Hall Renovation	95% Construction Documents Estimate	The Concord Group, Cost Consultant	1/6/2010	50KW	\$ 365,000	\$ 54,750.00	\$ 419,750.00	\$ 610,449.94	\$ 12.21	Seems much higher than today's technologies/costs. 2010 construction costs ended up being lower than 2009 due to the impacts of the financial crisis. Actual costs are not available since the PV system was designed as part of the project from the start. The costs were part of a lump-sum bid amount that included all electrical work for the building. The PV costs were not broken out in the bid as alternates. Therefore, the contractor's itemized costs for the PV system are not readily available.
Lincoln Hall Renovation	Executed Contract Value	Electrical Systems Inc.	6/18/2009	51.52KW			\$320,344.44	\$ 447,264.47	\$ 8.68	Seems much higher than today's technologies/costs. The PV system was added on after the main contracts were awarded. Therefore, a separate contract value was available for this work.

*** Note Soft Costs are excluded (Design costs, owner's costs, etc.)



Roof of EIB

Appendix B

Fuel Cell Calculations

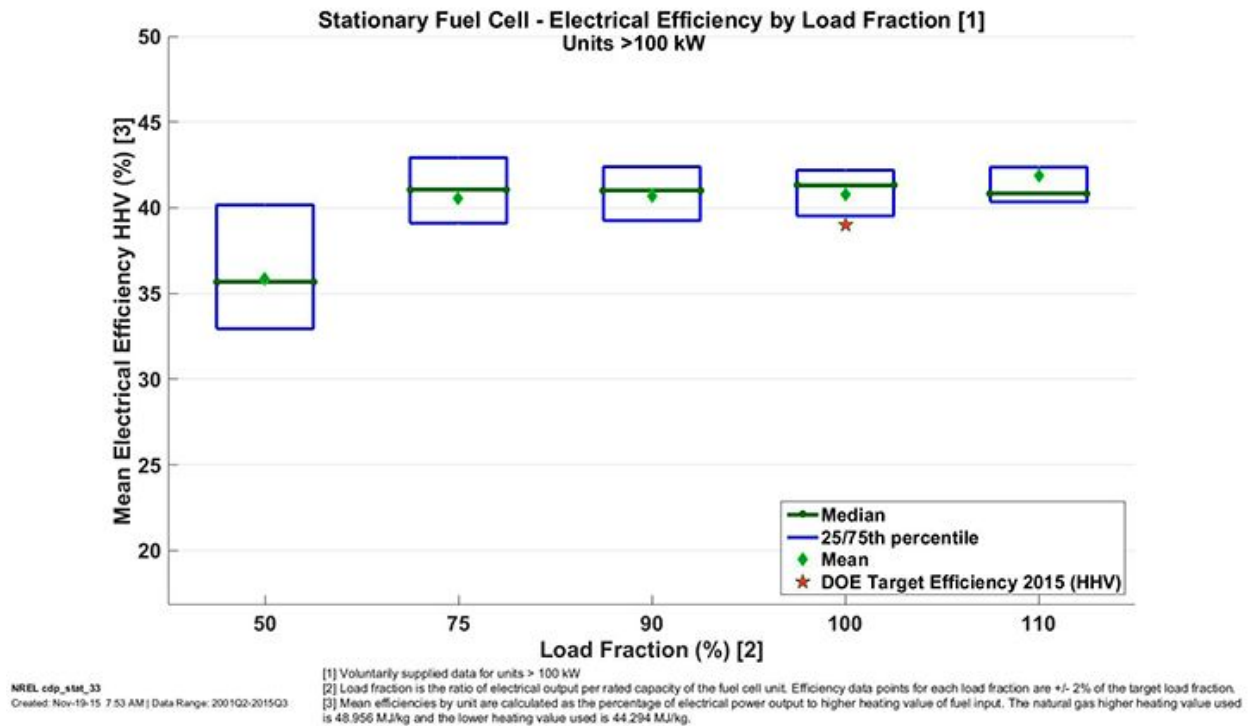


Figure 2: Efficiency [16]

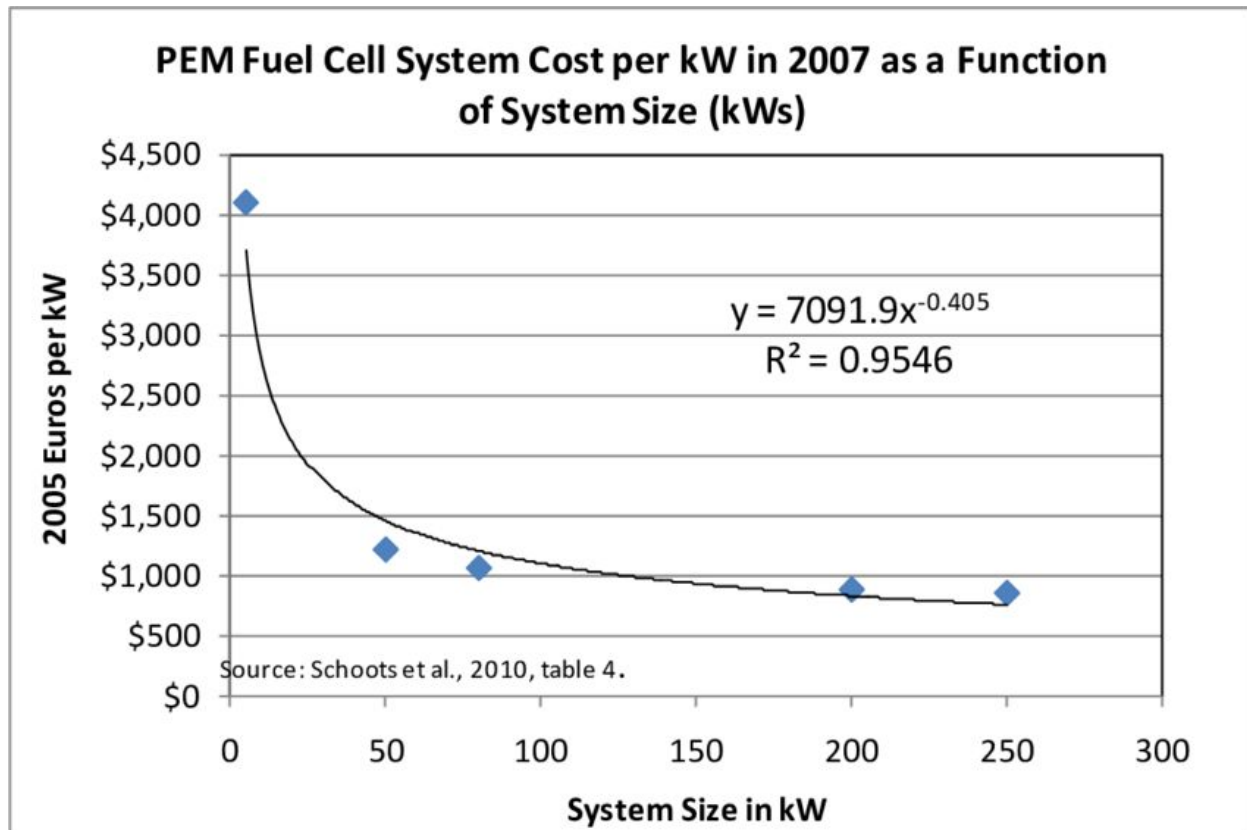


Figure 3: Fuel Cell Cost for 100 kW [17]

Table 1: Fuel Cell Cost

Peak Power of EIB (kW)	408
Efficiency of Fuel Cell	40%
Fuel Cell Power (kW)	1020
Number of 100 kW Fuel Cells	10
Number of 5 kW Fuel Cells	5
Cost of 5 kW Fuel Cell (Euros)	\$ 22,000.00
Cost of 100 kW Fuel Cell	\$ 121,922.98
Inflation Index 2007	167
Inflation Index 2019	289
Euro to Dollars	1.11
Fuel Cell Cost	\$ 2,262,212.11
Platinum Cost (per kW)	\$ 8.00
Inflation Index 2019	289
Inflation Index 2012	200
Total Cost of Platinum	\$ 6,813.60
Hydrogen for 5 kW (kg)	2
Hydrogen for 1020 kW (kg)	408
Backup Hydrogen (kg)	408
Price of Hydrogen (per kg)	\$ 8.00
Total Hydrogen Cost	\$ 6,528.00
Hydrogen Storage Price (for 18 kg)	\$ 50,000.00
Inflation Index 2007	167
Inflation Index 2019	289
Total Hydrogen Storage Cost	\$ 3,922,554.89
Total Upfront Fuel Cell Cost	\$ 6,184,767.00

Table 2: Diesel Generator Calculations

Diesel Generator 500	\$ 93,571
Capacity (kW)	500
# of generators	1
Permitting and Installation Costs for 2	\$ 24,000
Installation Costs (Scaled)	\$ 480,000
Fuel	
Fuel Consumed (liters per hour)	112.5
Fuel Cost (\$/liter)	\$0.79
Fuel Cost per year	\$1,600
Fuel cost for 15 years	\$23,996
Fuel Delivery Cost per liter (13% of cost)	\$0.10
Fuel Delivery Cost for 15 years	\$ 3,120
Fuel Tank Storage	
Fuel consumed per year (liters)	2,025
Fuel consumed per year (gallons)	535
Fuel Tank Storage capacity (gallon)	285
Tank Cost	1,100
Cost for 2 Fuel Storage Tanks	\$ 2,200
Maintenance	
Air Filter Cost (Change every 2 years)	\$ 200
Total Air Filter Cost (12x over 15 years)	\$ 1,500
Oil consumed (quarts)	174
Cost of Oil Change per quart	\$ 3
Total Oil Cost (1x per year) x 15 years	\$ 7,830
Oil Filter Cost	\$ 45
Oil Filter Cost (1x per year) x 15 Years	\$675
Coolant (gallon) (every 4 years or 6000h)	80
Cost of Coolant (\$/gallon)	\$ 6
Total Coolant price (3x over 15 years)	\$ 1,440
Fan Belt (change every 6 years)	\$ 300
Total Fan Belt Cost (2x over 15 years)	\$ 600
Battery (change every three years)	\$ 1,600
Total Battery cost (5x of 15 years)	\$ 8,000
15-year Total Maintenance Cost =	\$ 20,045
15-year Cost Diesel Generator System =	\$ 622,932

Table 3: Photovoltaic Analysis Calculations

Peak Hours (hr)	4.4		Battery Analysis	
Energy of Solar Panel (kWh/ 469solar panel)	446.6		Cost of Battery (\$/kWh)	250
Solar Panel Length (ft)	5.25		Energy Used	2,448.00
Solar Panel Width (ft)	3.083		Cost of the Batteries	\$ 612,000.00
Amount of Solar Panels Available	469		Cost of Batteries for LifeCycle	\$ 2,448,000.00
Area required for Solar Panels (ft^2)	7,591			
Peak Power Usage (kW)	408			
Available Area of Roof Space on EIB (ft^2)	9,856		Inefficiency factor	1.05
Total Energy Used (kWh)	1589		80% Depth Discharge	1.2
Total Energy Produced by Solar Panel (kWh)	1408		Hours of Storage	6
Number of Solar Panels	1.128551136		Battery Bank Sizing (kW)	3084.48
Cost of Solar Panels (\$/kW)	\$ 350.00			
Cost of Solar Panels	\$ 142,800.00			
80A 600 VDC Charge controller			Total Cost Upfront:	\$ 908,896.12
Charge Controller (\$)	1,347.16		Total Cost 15 years:	\$ 2,744,896.12
Amps	8500			
Amount of Charge controllers	106.25			
Cost of Charge Controller	\$ 144,146.12			
Inverter (\$/kW)	\$ 250.00			
Cost of Inverter	\$ 102,000.00			
Installation (\$/W)	0.5			
Cost of Installation	\$ 50,750.00			